

GITHUB WEEK - 1 ASSIGNMENT

Cybersecurity Asset Inventory System

Problem Statement

An organization maintains several IT assets such as computers, servers, routers, switches, and software applications. Managing these assets manually makes it difficult to identify the assets, track their security status, and determine which assets require immediate attention.

Develop a **Cybersecurity Asset Inventory System** that allows a security administrator to **add, search, update, delete, and display** information about the organization's IT assets. The system should classify each asset based on its **asset type** and **security risk level**.

1. Objective

To develop a **Cybersecurity Asset Inventory System** that manages information about an organization's IT assets and helps identify assets based on their **type, risk level, and security status**.

2. Problem Description

The system maintains information about different IT assets in an organization. For each asset, it accepts:

- Asset ID

- Asset Name
- Asset Type
- IP Address
- Operating System
- Owner/Department
- Risk Level
- Security Status

The system should allow the security administrator to add, search, update, delete, and display asset information, and classify assets according to their **Asset Type, Risk Level, and Security Status**.

3. Asset Type

- Workstation
- Server
- Router
- Switch
- Application

4. Risk Level

- Low
- Medium
- High
- Critical

5. Security Status

- Secure
- Warning
- Vulnerable

6. Input

The system accepts the following information for each asset. The number of assets is also entered by the user.

```
Asset ID
Asset Name
Asset Type
IP Address
Operating System
Owner/Department
Risk Level
Security Status
```

7. Sample Input

```
Enter number of assets: 3
```

```
Asset 1
Asset ID: A101
Asset Name: HR-PC-01
Asset Type: Workstation
IP Address: 192.168.1.10
Operating System: Windows 11
Department: HR
Risk Level: Medium
Security Status: Secure
```

```
Asset 2
Asset ID: A102
Asset Name: Web-Server
Asset Type: Server
IP Address: 192.168.1.20
Operating System: Ubuntu
Department: IT
```

Risk Level: Critical
Security Status: Vulnerable

Asset 3
Asset ID: A103
Asset Name: Core-Router
Asset Type: Router
IP Address: 192.168.1.1
Operating System: Cisco IOS
Department: Network
Risk Level: High
Security Status: Warning

8. Algorithm

1. Start the program.
2. Enter the number of assets.
3. Create a structure to store the asset information.
4. Get the Asset ID from the user.
5. Get the Asset Name.
6. Get the Asset Type.
7. Get the IP Address.
8. Get the Operating System.
9. Get the Owner/Department.
10. Get the Risk Level.
11. Get the Security Status.
12. Store the asset information.
13. Repeat the process for all assets.
14. Display the stored asset information.
15. Classify the assets according to their risk level and security status.
16. Display the total number of assets.
17. Display the number of critical, high-risk, medium-risk, and vulnerable assets.
18. Stop the program.

9. Program

File Name: asset_inventory.py

```
# Cybersecurity Asset Inventory System

assets = []

n = int(input("Enter number of assets: "))

for i in range(n):
    print("\nAsset", i + 1)
```

```

        asset_id = input("Asset ID: ")    asset_name
= input("Asset Name: ")    asset_type =
input("Asset Type: ")    ip_address = input("IP
Address: ")    operating_system =
input("Operating System: ")    department =
input("Department: ")    risk_level = input("Risk
Level: ")    security_status = input("Security
Status: ")

        asset = {
"asset_id": asset_id,
        "name": asset_name,
        "type": asset_type,
        "ip": ip_address,
        "os": operating_system,
        "department": department,
        "risk": risk_level,
        "status": security_status
}

        assets.append(asset) print("\n=====")
print("CYBERSECURITY ASSET INVENTORY")
print("=====")

for asset in assets:
    print("Asset ID :", asset["asset_id"])    print("Asset
Name :", asset["name"])    print("Asset Type :",
asset["type"])    print("IP Address :", asset["ip"])
print("OS :", asset["os"])    print("Department :",
asset["department"])    print("Risk Level :",
asset["risk"])    print("Status :", asset["status"])
print("-----")

critical = sum(1 for asset in assets if asset["risk"] == "Critical") high
= sum(1 for asset in assets if asset["risk"] == "High") medium = sum(1 for
asset in assets if asset["risk"] == "Medium") vulnerable = sum(1 for asset
in assets if asset["status"] == "Vulnerable")

print("Total Assets :", len(assets))
print("Critical Assets :", critical)
print("High Risk Assets :", high)
print("Medium Risk Assets :", medium)
print("Vulnerable Assets :", vulnerable)

print("=====")

```

10. Program Explanation

The program first creates an empty list to store the information about the organization's

```
assets = []
```

The number of assets is obtained from the user.

```
n = int(input("Enter number of assets: "))
```

A for loop is used to collect the information for each asset.

```
for i in range(n):
```

For every asset, the program collects:

```
asset_id = input("Asset ID: ") asset_name =  
input("Asset Name: ") asset_type =  
input("Asset Type: ") ip_address = input("IP  
Address: ") operating_system =  
input("Operating System: ") department =  
input("Department: ") risk_level = input("Risk  
Level: ") security_status = input("Security  
Status: ")
```

The collected information is stored in a dictionary.

```
asset = {      "id":  
asset_id,  
    "name": asset_name,  
    "type": asset_type,  
    "ip": ip_address,  
    "os": operating_system,  
    "department": department,  
    "risk": risk_level,  
    "status": security_status  
}
```

The asset is then added to the asset list.

```
assets.append(asset)
```

After collecting all assets, the program displays the **Cybersecurity Asset Inventory**. It also calculates the number of total, critical, high-risk, medium-risk, and vulnerable assets.

11. Sample Output

```
=====
                CYBERSECURITY ASSET INVENTORY
=====

Asset ID : A101
Asset Name : HR-PC-01
Asset Type : Workstation
IP Address : 192.168.1.10
OS : Windows 11
Department : HR
Risk Level : Medium
Status : Secure
-----

Asset ID : A102
Asset Name : Web-Server
Asset Type : Server
```

```

IP Address : 192.168.1.20
OS : Ubuntu
Department : IT
Risk Level : Critical
Status : Vulnerable

```

```

-----
Asset ID : A103
Asset Name : Core-Router
Asset Type : Router
IP Address : 192.168.1.1
OS : Cisco IOS
Department : Network
Risk Level : High
Status : Warning

```

```

-----
Total Assets : 3
Critical Assets : 1
High Risk Assets : 1
Medium Risk Assets : 1
Vulnerable Assets : 1
=====

```

12. Output Screenshot

```

File Edit Selection View Go Run Terminal Help
asset_inventory.py x
Python3 asset_inventory.py
100 print("\n=====")
101 print("CYBERSECURITY ASSET INVENTORY")
102 print("=====")
103 for asset in assets:
104     print("Asset ID :", asset["id"])
105     print("Asset Name :", asset["name"])
106     print("Asset Type :", asset["type"])
107     print("IP Address :", asset["ip"])
108     print("OS :", asset["os"])
109     print("Department :", asset["department"])
110     print("Risk Level :", asset["risk"])
111     print("Status :", asset["status"])
112     print("-----")
113
114 critical = sum(1 for asset in assets if asset["risk"] == "Critical")
115 high = sum(1 for asset in assets if asset["risk"] == "High")
116 medium = sum(1 for asset in assets if asset["risk"] == "Medium")
117 vulnerable = sum(1 for asset in assets if asset["status"] == "Vulnerable")
118
119 print("Total Assets :", len(assets))
120 print("Critical Assets :", critical)
121 print("High Risk Assets :", high)
122 print("Medium Risk Assets :", medium)
123 print("Vulnerable Assets :", vulnerable)
124 print("=====")
125
126 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
127
128 PS C:\Users\ifdule\Week-1> python asset_inventory.py
129 Enter number of assets: 3
130
131 Asset 1:
132 Asset ID: A101
133 Asset Name: WEBC-01
134 Asset Type: WebApplication
135 IP Address: 192.168.1.10
136 Operating System: Windows 11
137 OS: Windows
138 Risk Level: Medium
139 Security Status: Secure
140
141 Asset 2: A102
142 Asset Name: Web-Server
143 Asset Type: Server
144 IP Address: 192.168.1.20
145 Operating System: Ubuntu
146 Department: IT
147 Risk Level: Critical
148 Security Status: Vulnerable
149
150 Asset 3: A103
151 Asset Name: Core-Router
152 Asset Type: Router
153 IP Address: 192.168.1.1
154 Operating System: Cisco IOS
155 Department: Network
156 Risk Level: High
157 Security Status: Warning
158
159 =====
160 CYBERSECURITY ASSET INVENTORY
161 =====
162 Asset ID : A101
163 Asset Name : WEBC-01
164 Asset Type : WebApplication
165 IP Address : 192.168.1.10
166 OS : Windows
167 Department : IT
168 Risk Level : Medium
169 Status : Secure
170 -----
171 Asset ID : A102
172 Asset Name : Web-Server
173 Asset Type : Server
174 IP Address : 192.168.1.20
175 OS : Ubuntu
176 Department : IT
177 Risk Level : Critical
178 Status : Vulnerable
179 -----
180 Asset ID : A103
181 Asset Name : Core-Router
182 Asset Type : Router
183 IP Address : 192.168.1.1
184 OS : Cisco IOS
185 Department : Network
186 Risk Level : High
187 Status : Warning
188 -----
189 Total Assets : 3
190 Critical Assets : 1
191 High Risk Assets : 1
192 Medium Risk Assets : 1
193 Vulnerable Assets : 1
194 =====

```

13. GitHub Repository Structure

```

Week-1/
|-- asset_inventory.py
|-- README.md
`-- output.png

```

asset_inventory.py - Python source code

README.md - Project description and instructions output.png

- Screenshot of the program output

14. GitHub Submission Steps

Step 1 - Create the Python file: Create `asset_inventory.py` and paste the program into it.

Step 2 - Execute the program: Run it and enter the sample data for 3 assets. The final output shows Total 3, Critical 1, High Risk 1, Medium Risk 1, Vulnerable 1.

Step 3 - Take screenshot: Capture the complete execution and output and save it as `output.png`.

Step 4 - Create README: Create `README.md` with the project description.

Step 5 - Upload to GitHub: Upload `asset_inventory.py`, `README.md` and `output.png` to the Week-1 repository.

15. Git Commands

```
git add asset_inventory.py README.md output.png git commit -m  
"Added Week 1 Cybersecurity Asset Inventory System" git push
```

For a new local repository:

```
git init git add . git commit -m "Added Week 1 Cybersecurity  
Asset Inventory System" git push
```

16. GitHub Repository Verification

After pushing, verify that the repository contains:

```
asset_inventory.py  
README.md  
output.png
```

- Open `asset_inventory.py` on GitHub and check the source code displays correctly.
- Open `README.md` and check the description, code, and output information display correctly.
- Check that `output.png` was uploaded successfully.

17. Result

The **Cybersecurity Asset Inventory System** was successfully implemented and executed to store and display information about organizational IT assets. For the sample input containing 3 assets, the program displayed the asset information and calculated:

```
Total Assets : 3  
Critical Assets : 1  
High Risk Assets : 1  
Medium Risk Assets : 1  
Vulnerable Assets : 1
```

The Python program, README file, and output screenshot were uploaded to the GitHub repository successfully.

18. Submission Checklist

- ☐ asset_inventory.py created
- ☐ Python program executed successfully
- ☐ 3 sample assets entered
- ☐ Correct output obtained
- ☐ Output screenshot taken
- ☐ output.png added
- ☐ README.md created
- ☐ All three files uploaded to GitHub
- ☐ GitHub repository link copied
- ☐ PDF/report completed
- ☐ Output screenshot attached in the PDF

19. GitHub Repository Link

GitHub Repository: [[Paste your GitHub repository link here](#)]

Example: `https://github.com/your-username/week-1`